

HOMEWORK 1, SOLUTIONS

(10 points) 1. Prove that $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

Solution. Let $x \in A \setminus (B \cap C)$. This means that $x \in A$, and $x \notin B \cap C$. Assume first that $x \notin B$. Then we have $x \in A \setminus B$, so in particular, $x \in (A \setminus B) \cup (A \setminus C)$.

Assume on the other hand that $x \in B$. Then, as $x \notin B \cap C$, we conclude that $x \notin C$. But then $x \in A \setminus C$, so in particular, $x \in (A \setminus B) \cup (A \setminus C)$.

This proves that $A \setminus (B \cap C) \subset (A \setminus B) \cup (A \setminus C)$. We now prove the converse. Let $x \in (A \setminus B) \cup (A \setminus C)$. Then either $x \in A \setminus B$, or $x \in A \setminus C$. Suppose $x \in A \setminus B$ first. As $B \cap C \subset B$, then in particular $x \in A \setminus (B \cap C)$ as well.

The case $x \in A \setminus C$ is similar. In conclusion, $(A \setminus B) \cup (A \setminus C) \subset A \setminus (B \cap C)$ as well. This proves the result.

(10 points) 2. Use induction to prove that for all $n \geq 2$

$$\left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{9}\right) \left(1 - \frac{1}{16}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}.$$

Solution. For $n = 2$ we need $1 - \frac{1}{4} = \frac{3}{4}$, which is true. Assume that

$$(1) \quad \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{9}\right) \left(1 - \frac{1}{16}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}.$$

holds and want to prove that

$$(2) \quad \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{9}\right) \left(1 - \frac{1}{16}\right) \dots \left(1 - \frac{1}{(n+1)^2}\right) = \frac{(n+1)+1}{2(n+1)}.$$

Indeed, we have

$$\begin{aligned} & \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{9}\right) \left(1 - \frac{1}{16}\right) \dots \left(1 - \frac{1}{(n+1)^2}\right) \\ &= \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{9}\right) \left(1 - \frac{1}{16}\right) \dots \left(1 - \frac{1}{n^2}\right) \left(1 - \frac{1}{(n+1)^2}\right) \\ &= \frac{n+1}{2n} \left(1 - \frac{1}{(n+1)^2}\right), \end{aligned}$$

where in the last line we used (1). Now we have

$$\begin{aligned} \frac{n+1}{2n} \left(1 - \frac{1}{(n+1)^2}\right) &= \frac{n+1}{2n} \frac{n(n+2)}{(n+1)^2} \\ &= \frac{n+2}{2(n+1)}, \end{aligned}$$

which proves (2). This completes the induction proof.

(10 points) 3. Use induction to prove that

$$(2n + 1) + (2n + 3) + (2n + 5) + \dots + (4n - 1) = 3n^2$$

for all $n \geq 1$.

Solution. For $n = 1$ the identity is true as we have $3 = 3$. We now assume that the identity is true for n , that is

$$(3) \quad (2n + 1) + (2n + 3) + (2n + 5) + \dots + (4n - 1) = 3n^2$$

and prove it for $n + 1$. Replacing n by $n + 1$ everywhere in (3), the statement we need to prove is

$$(2(n + 1) + 1) + (2(n + 1) + 3) + (2(n + 1) + 5) + \dots + (4(n + 1) - 1) = 3(n + 1)^2,$$

which is the same as

$$(4) \quad (2n + 3) + (2n + 5) + \dots + (4n - 1) + (4n + 1) + (4n + 3) = 3(n + 1)^2$$

We use (3) to compute the left side of (4) as follows:

$$\begin{aligned} & (2n + 3) + (2n + 5) + \dots + (4n - 1) + (4n + 1) + (4n + 3) \\ &= ((2n + 1) + (2n + 3) + \dots + (4n - 1)) \\ & \quad + (4n + 1) + (4n + 3) - (2n + 1) \\ &= 3n^2 + (4n + 1) + (4n + 3) - (2n + 1) \\ &= 3n^2 + 6n + 3 \\ &= 3(n + 1)^2. \end{aligned}$$

This proves (4) and completes the induction proof.

(10 points) 4. If x is rational, $x \neq 0$, and y irrational, prove that $x + y$ and xy are irrational. Is the sum of two irrational numbers always irrational?

Solution. The sum and product of two rational numbers are rational as well. So, if $z = x + y$ is rational, then $y = z - x$ is rational as well. This is a contradiction. Similarly, if $w = xy$ is rational, then $y = \frac{w}{x}$ is rational as well, which is a contradiction.

The sum of two irrational numbers may be rational, as it is the case with $x = \sqrt{2}$ and $y = -\sqrt{2}$.